

Computer Science & IT



Compiler Design



Intermediate Code Generation

Lecture No . 02



By- Ravindra Sir

Recap of Previous Lecture



Topic

Parsers

Topic

Topic

Topics to be Covered



Topic

Par SEM

Topic

Topic

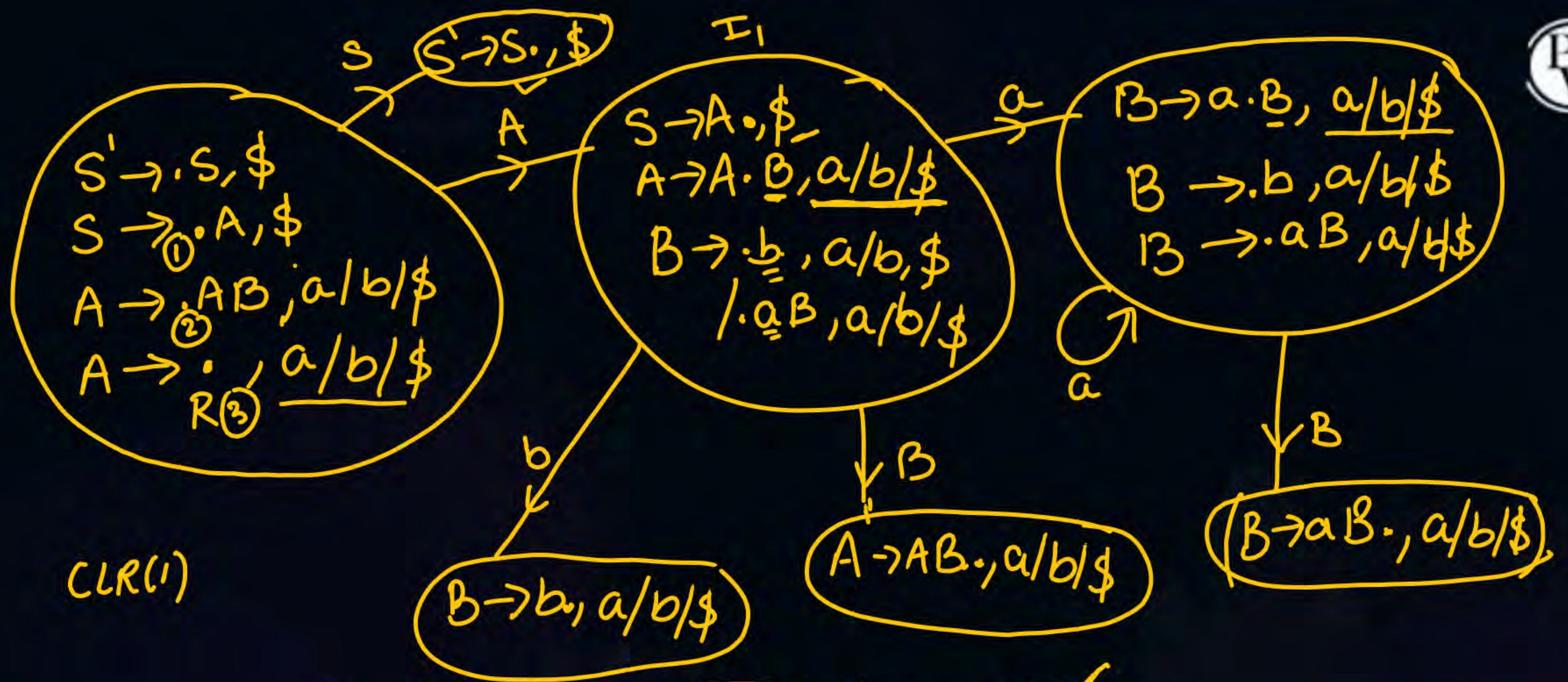
$S \rightarrow A$
 $A \rightarrow AB$
 $\quad / \epsilon$
 $B \rightarrow aB$
 $\quad / b$

$S' \rightarrow \cdot S, \$$
 $S \rightarrow \cdot A, \$$

$A \rightarrow \cdot A B, \$$
 $\quad / \cdot, \$$
 $A \rightarrow \cdot A B, a/b$
 $\quad / \cdot, a/b$

$\Rightarrow$

$S' \rightarrow \cdot S, \$$
 $S \rightarrow \cdot A, \$$
 $A \rightarrow \cdot A B, \$/a/b$
 $\quad / \cdot, \$/a/b$



CLR(1)

	$\checkmark a$	$\checkmark b$	$\checkmark \$$
I0	S	S	I1

CLR(1) = LAR(1) ✓

$$E \rightarrow E + T / T$$

$$T \rightarrow T * F / F$$

$$F \rightarrow id$$

LR(0) ✗
SLR(1)

$E \rightarrow E + T$
/ T
 $T \rightarrow T * F$
/ F
 $F \rightarrow id$

CLR(1)
LALR(1)

$E' \rightarrow \cdot E, \$$

$E \rightarrow \cdot E + T, \$$
/ $\cdot T$ / $\$$

$E \rightarrow \cdot E + T, +$
/ $\cdot T, +$

$E \rightarrow \cdot E + T, \$ / +$
/ $\cdot T, \$ / +$

$T \rightarrow \cdot T * F, \$ / +$
/ $\cdot F, \$ / +$
 $T \rightarrow \cdot T * F, *$
 $\cdot F / *$

$E' \rightarrow \cdot E \$$

$E \rightarrow \cdot E + T / \$, +$

$\cdot T / \$, +$

$T \rightarrow \cdot T * F / *, \$, +$

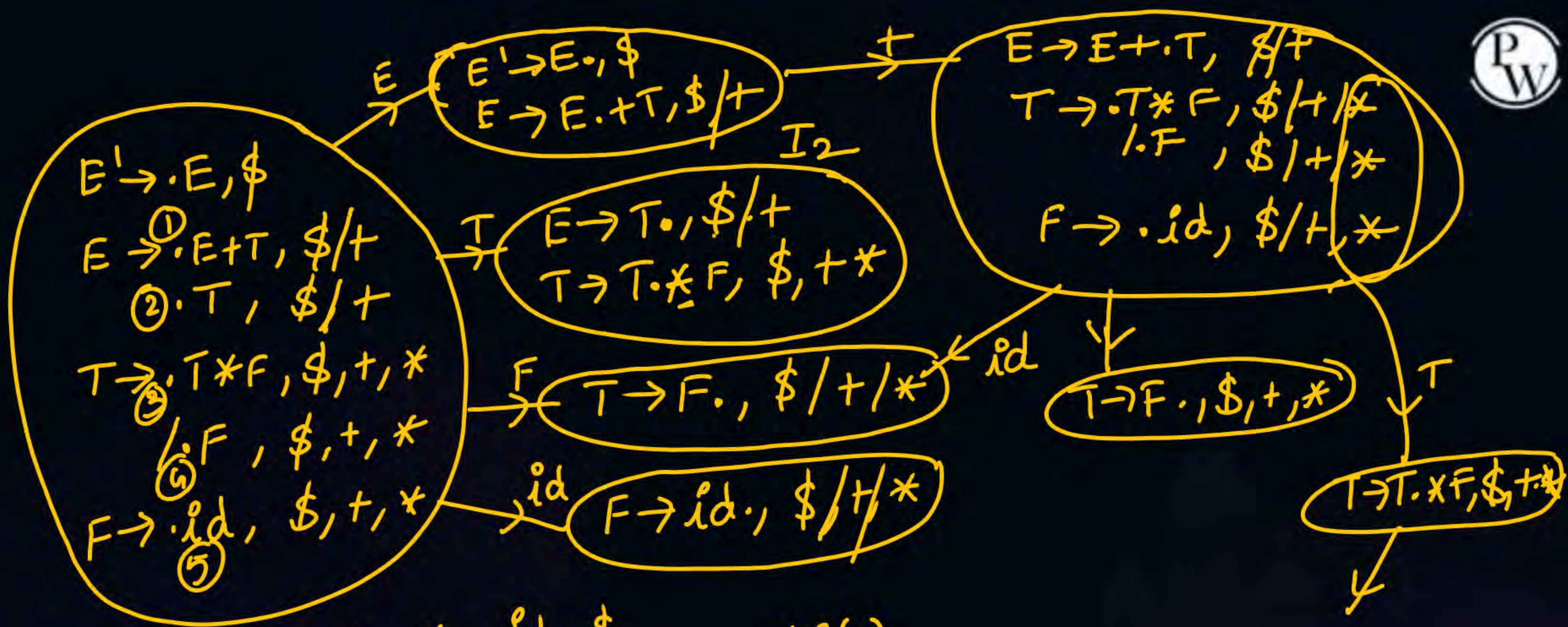
$T \rightarrow \cdot F / *, \$, +$

$F \rightarrow \cdot id / *, \$, +$

CLR(1)

LALR(1)

✓
✓



	$+$	$*$	id	$\$$
I_2	$\emptyset$	S		$\emptyset$

LR(1)



THANK - YOU